



Gesture-Controlled Presentation Tool



Project Overview

The **Gesture-Controlled Presentation Tool** is a computer vision–based system that enables **hands-free control of presentation slides** using real-time hand gesture recognition. It eliminates the need for physical devices such as remotes, keyboards, or mice and provides an intuitive, touchless way to interact with presentation software.

This project is especially useful in:

- Smart classrooms
- Academic presentations
- Hygiene-sensitive environments
- Situations where physical interaction is inconvenient

The system uses a webcam to detect hand movements and translates recognized gestures into slide navigation commands.



Key Objectives

- Enable **touchless slide navigation** using hand gestures
 - Provide **accurate and stable gesture recognition**
 - Prevent accidental slide changes using explicit activation
 - Offer a **popup gesture guide** for user assistance
 - Maintain compatibility with common presentation tools
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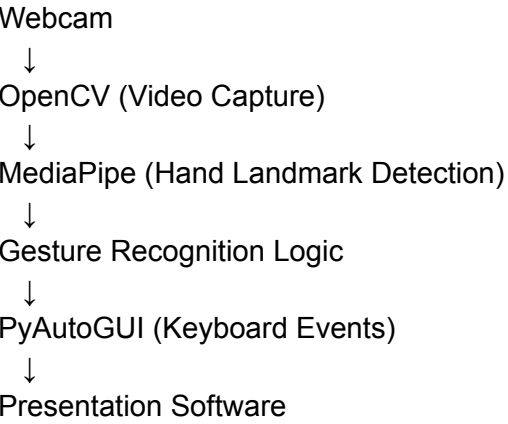


Technology Stack

Component	Description
Programming Language	Python 3.11

Computer Vision	OpenCV
Hand Tracking	MediaPipe Hands
System Control	PyAutoGUI
Hardware	Standard webcam

System Architecture



Supported Gestures & Actions

Gesture	Description	Action Performed
 Open Palm	Five fingers clearly visible	Activates gesture control
 Swipe Right	Rightward hand movement	Moves to next slide
 Swipe Left	Leftward hand movement	Moves to previous slide
 Fist	All fingers closed	Exits the slideshow
 Peace Sign	Two fingers raised	Displays popup gesture guide

Important:

Gesture actions work **only after control is activated using the open palm gesture**, ensuring stability and preventing unintended slide changes.

Gesture Guide Popup

- The **peace sign** (🙌) triggers a **floating popup window**
- The popup appears **beside the presentation**, not on top of it
- It displays a concise guide to all supported gestures
- The popup automatically disappears when the gesture is removed

This design provides user assistance without interrupting the presentation flow.

Gesture Recognition Logic

- Hand landmarks (21 key points) are detected using MediaPipe
 - Finger states are determined by comparing fingertip and joint positions
 - Swipe gestures are identified using horizontal landmark movement
 - A **cooldown timer** prevents repeated triggering
 - Only **one hand** is processed at a time for accuracy
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Project Folder Structure

```
gesture-control/
| — slide_control.py      # Final integrated implementation
| — webcam_test.py       # Webcam testing
| — hand_detect.py       # Hand landmark detection
| — finger_count.py      # Finger counting logic
| — requirements.txt     # Dependencies
| — README.md           # Project documentation
```

How to Run the Project

Create & Activate Virtual Environment

```
python3.11 -m venv gesture_env  
source gesture_env/bin/activate
```

2 Install Dependencies

```
pip install -r requirements.txt
```

3 Start Presentation

- Open **PowerPoint** or **Google Slides**
- Start slideshow in **fullscreen mode**
- Keep the presentation window in the foreground

4 Run the Application

```
python slide_control.py
```



System Requirements

- Python 3.11
 - Webcam (built-in or external)
 - Good lighting conditions
 - macOS / Windows / Linux
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Academic Explanation

“The system uses real-time hand landmark detection to recognize predefined gestures. These gestures are mapped to keyboard events using PyAutoGUI, allowing touchless control of presentation slides. An explicit activation gesture is used to avoid accidental slide transitions.”



Advantages

- Touchless and intuitive interaction
 - No additional hardware required
 - Works with existing presentation software
 - Reduces dependency on physical controllers
 - Easy to use and demo-friendly
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Future Enhancements

- Voice + gesture hybrid control
 - Annotation and highlighting gestures
 - Zoom and pointer gestures
 - Mobile application integration
 - AI-based adaptive gesture learning
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Conclusion

The Gesture-Controlled Presentation Tool demonstrates a practical application of **computer vision and human-computer interaction**.

By combining accuracy, usability, and thoughtful UI design, this project provides an effective solution for modern presentation environments.